Authorization & Audit Module for Autonomous Spectrum Coordination

A named human's signed "yes" before an AI agent does anything irreversible — with a receipt anyone verifies offline.

- For Spectrum Strike: the **governance & audit layer** under your triage agent.
- Open **Apache-2.0** standard + reference implementation. IETF Internet-Draft live on datatracker.

`agent → escalation decision → named-human signoff → authorization receipt → offline-verifiable`

Visual: *EP wordmark on dark field; pipeline line beneath; gold rule under the title.*

EMILIA Protocol, Inc. · Iman Schrock, Founder/PI · team@emiliaprotocol.ai

THE PROBLEM, IN DIU'S WORDS

What Spectrum Strike Is Asking For

SPEED

Collapse spectrum coordination & authorization from **90+ days** → **under 5 days** with AI agents.

HUMAN-ON-THE-LOOP

Triage **without persistent human-in-the-loop bottlenecking**, while keeping **human-on-the-loop** for flagged high-risk packets.

AUDITABLE GOVERNANCE

Auditable governance as policies change — the rules will move; the audit must hold.

SAFETY BAR

A hard **0% false-negative** bar on safety-critical packets.

Visual: *four-quadrant card; the two gold-bordered cards are EP's lane.*

THE GAP TODAY

Control & Accountability Are a Runtime Promise, Not a Verifiable Invariant

TODAY — INSIDE THE OPERATOR'S TRUST BOUNDARY

- "A human approved this" lives in a log the operator controls — **testimony, not evidence**.
- Logs can be edited, reconstructed, or lost; you trust the system that produced them.
- When policy changes mid-mission, nothing ties an old approval to the exact policy in force then.

What a 0%-false-negative bar actually needs

- An artifact that proves a **named** human approved that **exact** emission.
- Bound to the **exact policy** in force, **before** it executed.
- Checkable by a third party **without trusting the operator's runtime**.

Visual: *left, an editable "audit log" inside a dashed operator trust boundary; right (gold), a self-contained receipt that verifies outside it.*

EP IN ONE DIAGRAM

From Agent Judgment to Offline-Verifiable Proof

1 · PIP-007

Escalation decision

The agent's own signed record — "I judged this required human authorization." A claim by a party EP **identifies but never trusts**.

2 · SIGNOFF

Named-human signoff

WebAuthn device-bound approval over the exact action hash + pinned policy. The operator never holds the key.

3 · RECEIPT

Authorization receipt

Signoff + hash-pinned policy + append-only Merkle-anchored log, composed into one self-verifying document.

4 · VERIFY

Offline verification

Any third party validates with only the approver's public key — **no network**, no trust in the operator.

PIP-007 is the agent's **claim** that it escalated — bound into what the human signs, never a trust input, and never model introspection.

Visual: *left-to-right pipeline of four nodes with gold connectors; "no network required" tag under node 4.*

REQUIREMENT → PRIMITIVE MAP

Each Ask, Mapped to the Exact EP Primitive

DIU REQUIREMENT (RE-CONFIRM ON PORTAL)	EP PRIMITIVE	WHAT THE RECEIPT PROVES
Human-on-the-loop for flagged high-risk packets	**WebAuthn device-bound named-human signoff over the exact action hash**	A *named* human approved this *exact* spectrum release *before* it executed
Auditable governance *as policies change*	**Hash-pinned policy binding in every signoff**	Each approval is bound to the exact policy version — tamper-evident, replay-proof
Audit teeth for the 0% false-negative bar	**Offline-verifiable receipt + append-only Merkle-anchored log**	Any third party re-checks every release with only a public key, no operator trust
Why the agent asked for a human at all	**PIP-007 escalation attestation (signed, never trusted)**	The agent's stated escalation reason is bound into what the human signed
Disconnected / classified operation	**Air-gap installer (deploy/airgap/)**	Issue and verify with no network, forever

Visual: *the map itself — DIU asks in muted ink, EP primitives in gold.*

WHY DECISIVE FOR DOD

Built for Disconnected, Contested, Accountable Operations

DISCONNECTED

Receipts verify with **no network, forever**. The air-gap installer ships today (`deploy/airgap/`). No trust in, and no connection to, the operator's runtime — only the approver's public key.

TAMPER-EVIDENT

Hash-pinned policy means an old approval **can't be silently re-attributed** to a new rule. "Governance as policies change" stays honest — you can always show which rule authorized which release.

ATTRIBUTABLE

Every release **names the human** and carries the agent's stated escalation reason — discoverable in an after-action review, not buried in an operator-controlled log.

- This is what separates EP from a SaaS audit dashboard: offline by construction, in exactly the environments where connectivity and operator trust can't be assumed.

Visual: *three pillars — Disconnected · Tamper-evident · Attributable — over a faint "no network" globe.*

PROOF IT'S REAL

Formally Assured, Independently Verifiable, Installable Today

26	413,137	22	85	-01
TLA+ safety properties machine-checked	states explored	Alloy assertions, 0 counterexamples	red-team cases cataloged	IETF I-D revision, live on datatracker

- Independent **JS, Python, Go** verifiers + conformance suite — proven to agree on the adversarial vectors, every push.
- `npm i @emilia-protocol/verify` and check a real receipt in under a minute, offline.
- Scope honestly: the formal models prove the **authorization state machine**, not an AI model's behavior or a host's integrity.

Visual: *five-stat strip in gold (26 / 413,137 / 22 / 85 / -01), npm + datatracker line beneath.*

LIVE 60-SECOND DEMO

Issue a Receipt. Mutate One Field. Watch It Fail.

```
# issue a mock spectrum release (2401 MHz, AOR-7)
$ ep-issue issue --action action.json …
$ verify receipt.json --verification …

✅ VERIFIED — authorization receipt
✓ action_hash
✓ context_commitments
✓ signoff_signatures   ✓ sod
✓ inclusion   ✓ checkpoint_signature
✓ windows
attestation: present, consistent
```

```
# mutate ONE field: 2401 → 2455 MHz
# (adversary retargets the emission)
$ verify receipt-tampered.json …

⛔ NOT VERIFIED — authorization receipt
× action_hash
× context_commitments
✓ signoff_signatures   ✓ sod
× inclusion
✓ checkpoint_signature   ✓ windows
exit code 0 → 1
```

- The signature was over the **exact** action. Change the target, and the proof collapses — offline, with only a public key.

Visual: *split terminal, green "VERIFIED 7/7" vs. red "NOT VERIFIED ✕ action_hash". Full script in DEMO-SCRIPT.md; browser fallback at emiliaprotocol.ai/verify.*

A Non-Traditional Vendor, Built in the Open

- **EMILIA Protocol, Inc.** — Delaware C-corporation. Founder & PI: **Iman Schrock** (ORCID 0009-0004-0290-5433) · team@emiliaprotocol.ai.
- **Solo founder, pre-revenue, no outside capital** — stated plainly. DIU explicitly encourages small and non-traditional defense vendors; that is exactly who we are.
- Built in the open: Apache-2.0, a public PIP governance process, and a **three-author IETF collaboration** (EMILIA + PSEA / Yuthent + an independent verifier-model reviewer) coordinating the standard on the IETF secdispatch list.
- **Depth over headcount:** a formally-verified protocol, a three-language verifier suite, and a live reference deployment — shipped by a focused team-of-one with open-source contributors.

The mitigation for our size is structural: an open standard others can implement independently, and a teaming posture as the governance/audit module under a spectrum-AI prime.

Visual: *company + founder line; the three-author standards collaboration; "the team scales by adoption, not headcount."*

SOFTWARE MATURITY & PAST PERFORMANCE

Deployed and Independently Verifiable — TRL 6 Today, TRL 7 at MVP

DEPLOYED TODAY · TRL 6

npm-published verifiers, a live in-browser verifier, an MCP server, and a Class-A device-bound signoff exercised end-to-end on real hardware — the audit module validated in a relevant environment.

PATH TO TRL 7

The 07-10 MVP wires EP to a reference triage agent over MCP, issuing a receipt for every simulated high-risk release — demonstration in an operationally relevant Spectrum Strike context.

TRACK RECORD = THE ARTIFACT

No prior government contracts. Our past performance is **public and checkable**: IETF I-D -01, a formally-verified open-source protocol, npm packages, and a cross-language conformance suite proven to agree in CI.

- Stated honestly: **no production deployment at relying-party scale yet, and no DoD past performance.** The mitigation is verifiability — anyone can install the toolkit and confirm every claim in minutes.

Visual: *three pillars — Deployed (TRL 6) · Path to TRL 7 · Verifiable track record.*

A Thin Evidence Layer — Augments Systems of Record, Replaces Nothing

- **No migration.** The triage agent calls EP over a standard interface (REST API or MCP) at the moment of an irreversible release; EP returns a receipt keyed by `action_hash` + `receipt_id`, stored as a reference field alongside the existing record — a foreign key, not a rip-and-replace.
- **Three integration modes:** (1) inline API/MCP call before the write; (2) a sidecar that *refuses* an irreversible action without a valid receipt (`require-receipt`); (3) air-gapped issuance for disconnected or classified enclaves.
- **Open, documented interfaces** — Apache-2.0, OpenAPI, MCP. No proprietary connector and **no data egress**: the receipt verifies offline, so legacy federal data never leaves its enclave.
- **High-side ready:** stateless verification + the air-gap installer fit an IL5/IL6 enclave; the offline-verifiable receipt is the only artifact that needs to cross a boundary.

Visual: *a legacy system-of-record box with EP attached as a thin reference field; the receipt as the only artifact crossing the enclave boundary.*

HONEST SCOPE & TEAMING

What EP Is — and What It Needs From a Prime

EP — AUTHORIZATION & AUDIT LAYER

Signed escalation (PIP-007) · named-human signoff · hash-pinned policy · offline-verifiable receipts.

▲ MCP: triage agent calls EP before any irreversible release ▲

PRIME — SPECTRUM TRIAGE / RF MODELING / DECONFLICTION

EP is **not** the spectrum-parsing/triage engine. Best fit: teaming/sub as the "governance & audit module" under a spectrum-AI prime — or a standalone audit-module bid.

- **What EP needs from a prime:** the agent's escalation hook, the policy definitions to hash-pin, and named-approver enrollment.
- **A receipt does NOT prove:** the decision was wise/lawful; real-world biometric identity beyond key↔approver enrollment; or rendering faithfulness (crypto proves a key signed a hash — narrowed by structured rendering + signed display attestation, not eliminated).

Visual: *two stacked bands — gold "EP: authorization & audit" over muted "Prime: spectrum triage / RF / deconfliction" — MCP arrow between.*

ASK & ROADMAP

The Ask, the Timeline, and How to Reach Us

Advance EP as the governance/audit module for Spectrum Strike — as a sub to a spectrum-AI prime, or as a standalone audit-module evaluation.

Pitch · 2026-06-15

This deck + the 60-second offline issue→mutate→fail demo.

MVP · 2026-07-10

EP wired to a reference triage agent via MCP; receipts for every simulated high-risk release; offline audit bundle.

Live demo · 2026-08-25

End-to-end: agent escalates → named human signs off → receipts verify offline → a mutated release fails.

- **What we bring today:** TRL ~6 receipt/verify core · IETF I-D -01 · 3-language verifiers · air-gap installer. Pre-revenue, no customers, solo founder — stated plainly.

Contact: Iman Schrock · team@emiliaprotocol.ai · emiliaprotocol.ai · github.com/emiliaprotocol/emilia-protocol

Visual: *three-stop timeline (06-15 → 07-10 → 08-25) in gold with the contact line beneath.*